

ERA 5 calibration to Elexon power

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Overview

Updates

New

- Error diagnostics
- Spatiotemporal model for calibration

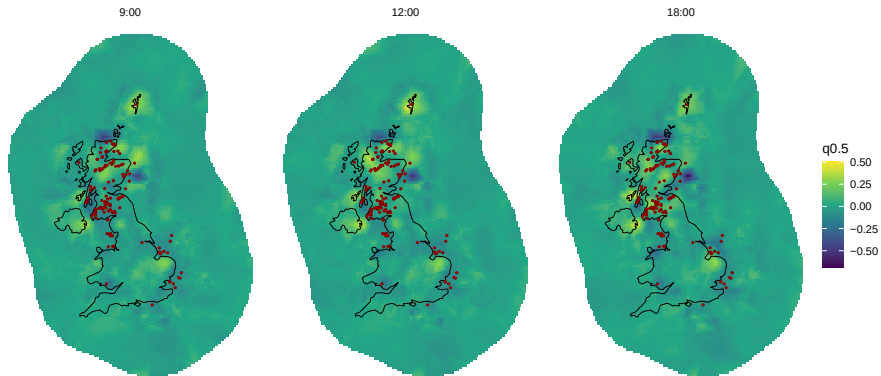
Spatiotemporal calibration model

Model overview

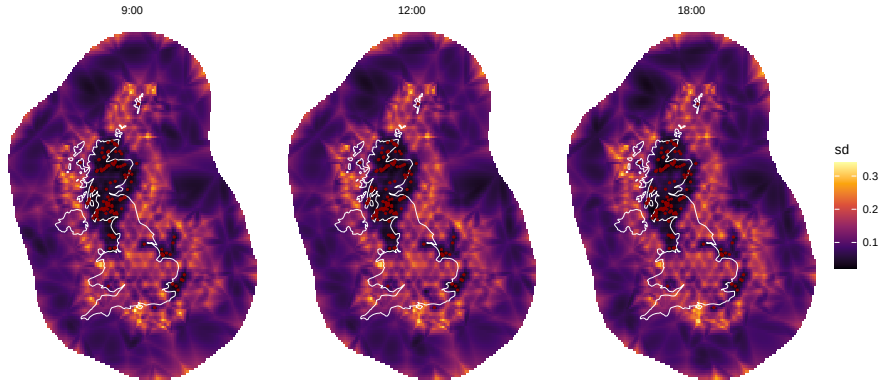
$$\mathbb{E}(p_{it}^{\text{calib}}) = \alpha + \beta_k + \beta_m + \beta_h + s_p(\hat{p}_{it}) + s_w(w_{it}) + u_{it}, \quad (1)$$

- Calibrated power: corrected output at site i , time t .
- α : intercept.
- Fixed effects $\beta_k, \beta_m, \beta_h$: location/season/time structure.
- ERA5 correction s_p : bias adjustment of ERA5 + generic PC estimate.
- Wind effect s_w : nonlinear wind–power correction.
- Spatiotemporal effect u_{it} : Spatial component + AR1 error.

Estimated spatial field



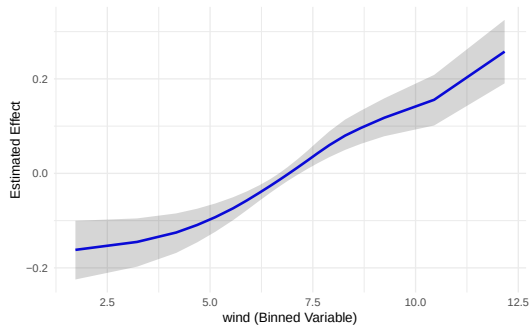
Standard deviation of the spatial field



Random Effects

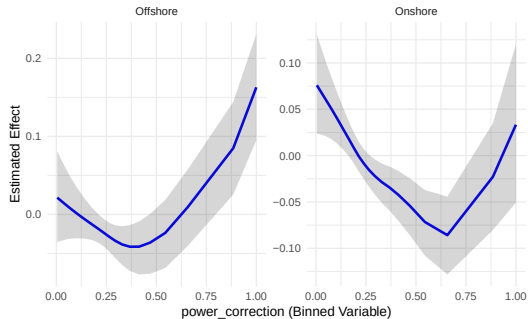
Correction by wind speed

Estimated effect for wind



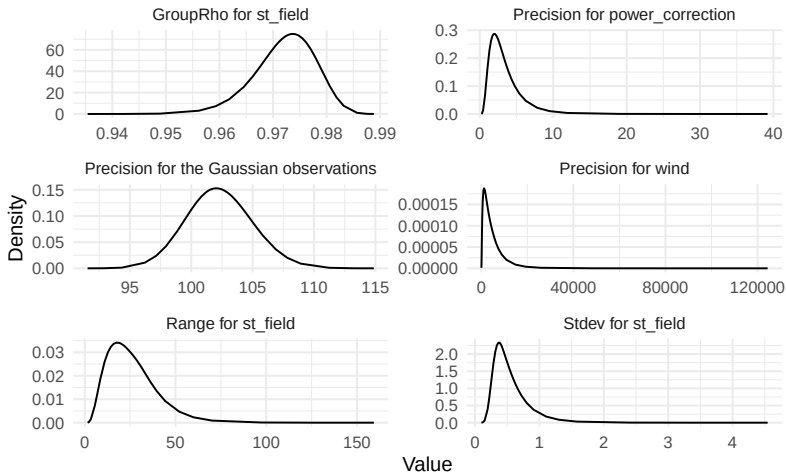
Correction by power regime

Estimated effect for power_correction



Hyperparameters

Density of Hyperparameters



Comparison scores

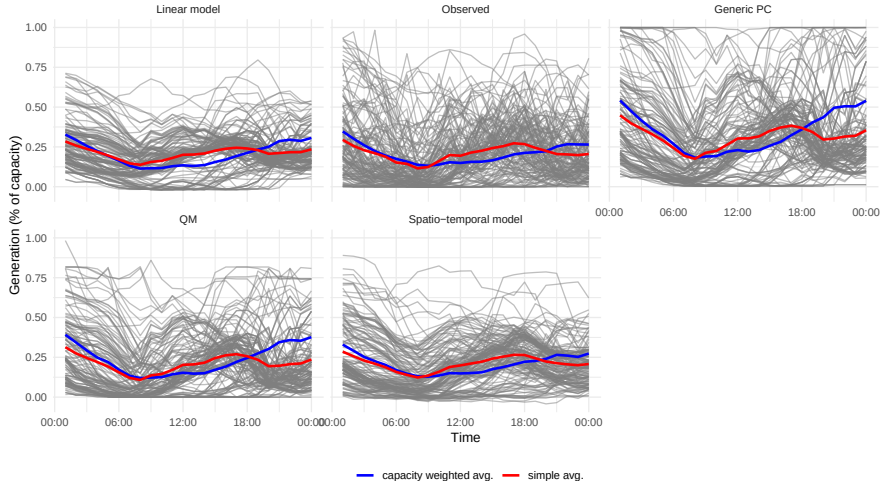
model	RMSE	MAE	Bias
Generic PC	0.213	0.153	1.02e-01
QM	0.154	0.108	-2.50e-04
Linear model	0.137	0.101	-1.15e-17
Spatio-temporal model	0.094	0.065	8.57e-11

Scores by Offshore / Onshore

tech_typ	model	RMSE	MAE	Bias
Wind Offshore	Generic PC	0.240	0.161	1.44e-01
Wind Offshore	QM	0.142	0.087	4.60e-02
Wind Offshore	Linear model	0.112	0.077	4.33e-16
Wind Offshore	Spatio-temporal model	0.065	0.040	-3.78e-03
Wind Onshore	Generic PC	0.204	0.151	8.77e-02
Wind Onshore	QM	0.159	0.114	-1.58e-02
Wind Onshore	Linear model	0.144	0.109	-1.61e-16
Wind Onshore	Spatio-temporal model	0.101	0.073	1.27e-03

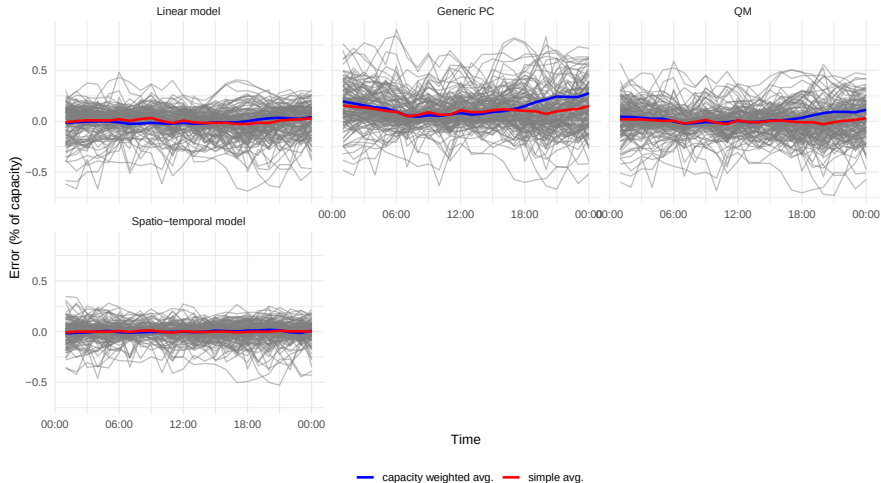
Estimates by hour

Power estimates Time Series 2025-08-10



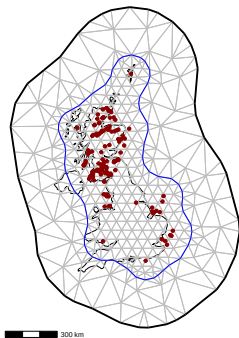
Errors by hour

Power estimates Time Series 2025-08-10

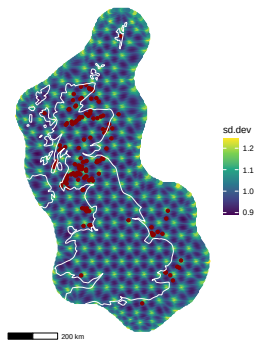


Model Mesh

Mesh used for spatial calibration model



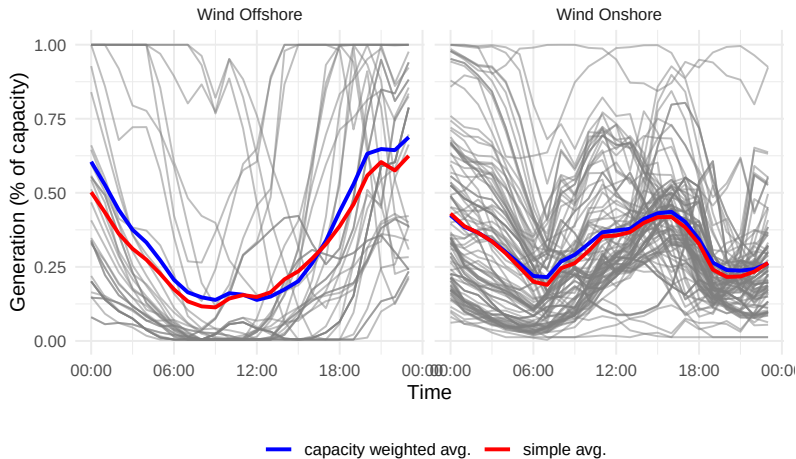
Mesh assessment



Time series of data and generic PC estimate

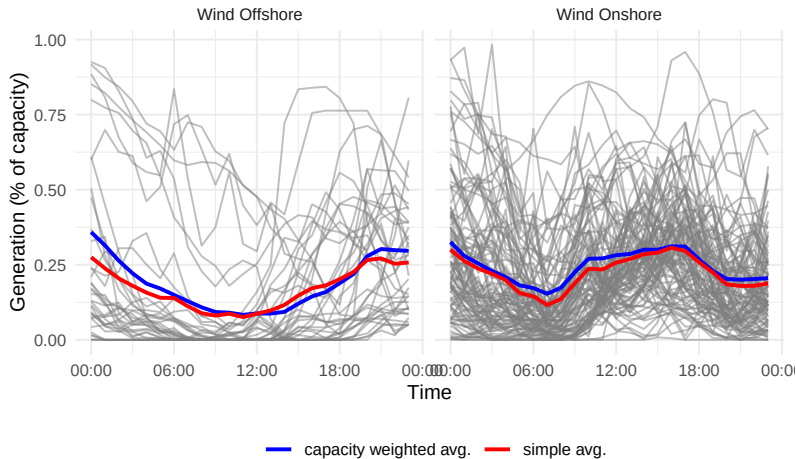
ERA5 data and generic PC estimate

ERA5 + Generic PC Time Series 2025-08-10

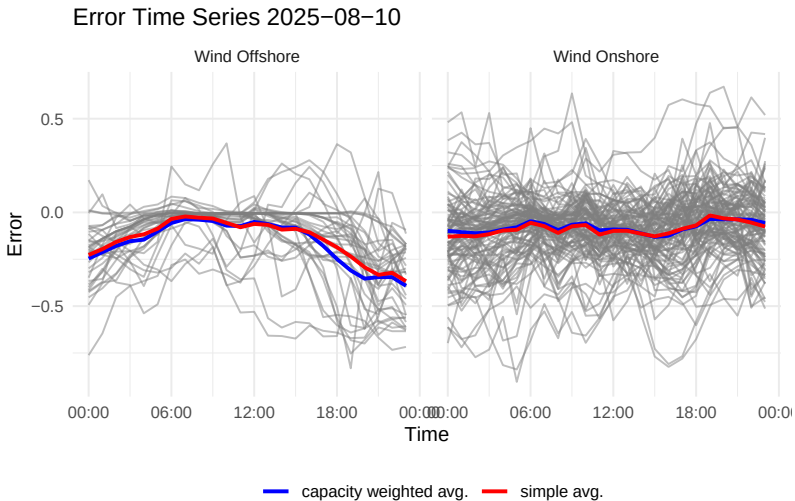


Observed data

Observed power Time Series 2025-08-10



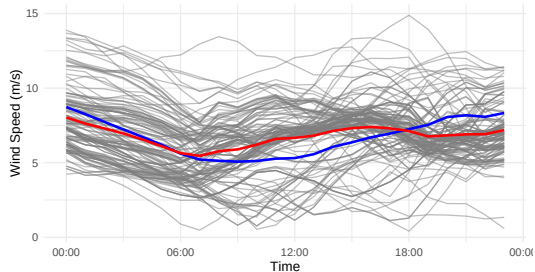
Error of generic PC estimate



ERA5 wind speed

Wind speed at wind farm locations

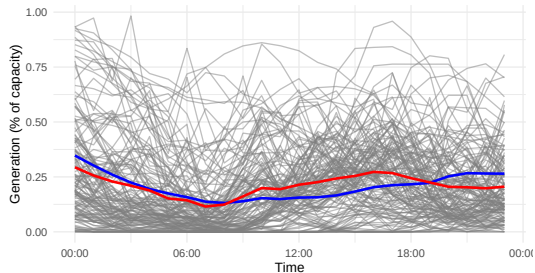
Wind Speed Time Series 2025-08-10



— capacity weighted avg. — simple avg.

Observed generation

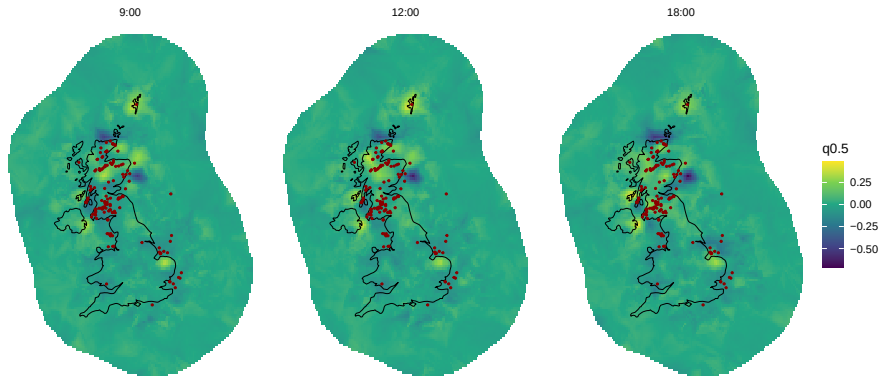
Observed power Time Series 2025-08-10



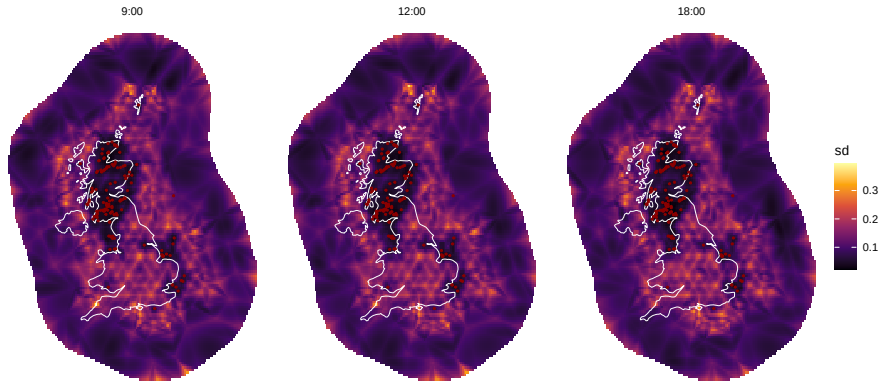
— capacity weighted avg. — simple avg.

Same day previous year

Estimated spatial field



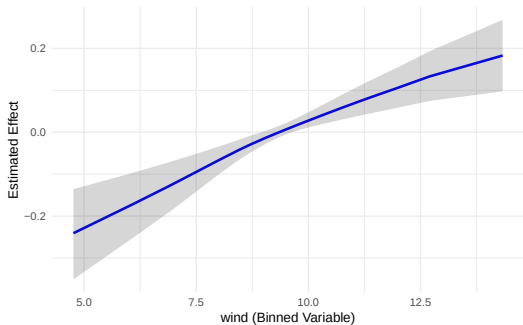
Standard deviation of the spatial field



Random Effects

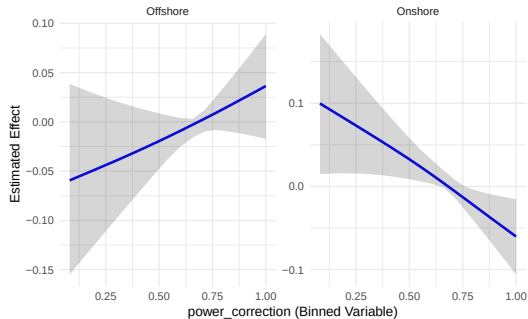
Correction by wind speed

Estimated effect for wind



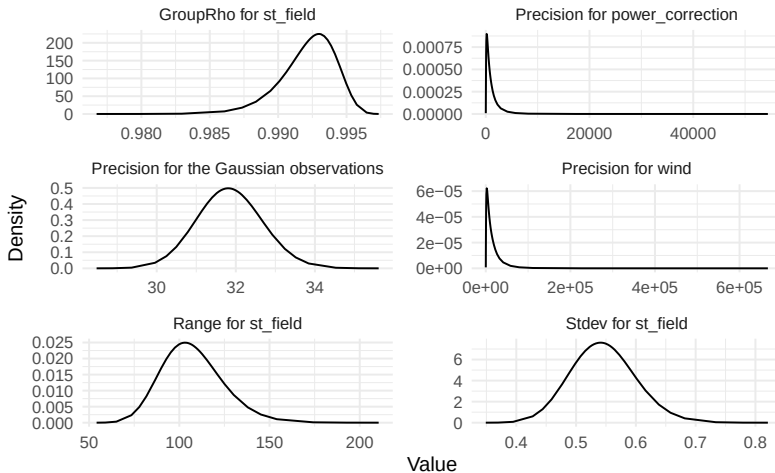
Correction by power regime

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Hyperparameters

Density of Hyperparameters



Comparison scores

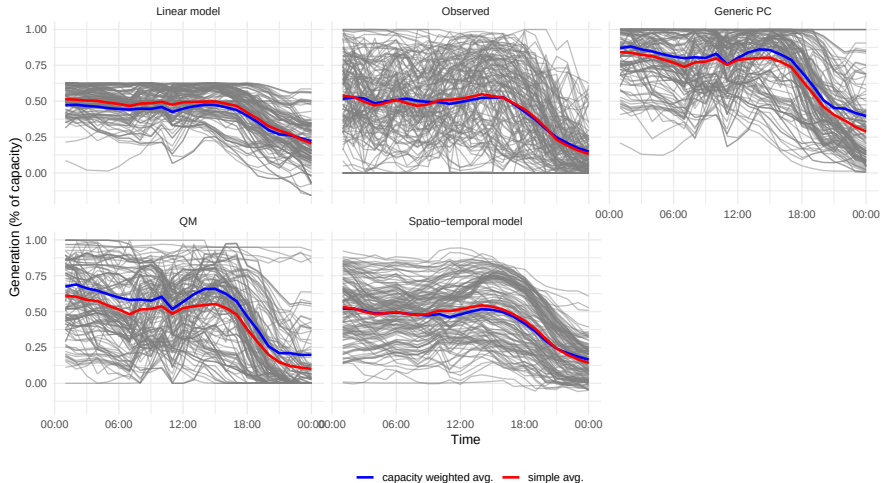
model	RMSE	MAE	Bias
Generic PC	0.385	0.299	2.50e-01
QM	0.314	0.239	-1.23e-04
Linear model	0.251	0.199	3.38e-17
Spatio-temporal model	0.170	0.125	1.08e-09

Scores by Offshore / Onshore

tech_typ	model	RMSE	MAE	Bias
Wind Offshore	Generic PC	0.449	0.369	3.61e-01
Wind Offshore	QM	0.341	0.236	1.47e-01
Wind Offshore	Linear model	0.211	0.162	7.09e-17
Wind Offshore	Spatio-temporal model	0.098	0.072	-4.90e-04
Wind Onshore	Generic PC	0.361	0.276	2.13e-01
Wind Onshore	QM	0.305	0.240	-4.88e-02
Wind Onshore	Linear model	0.263	0.211	2.16e-17
Wind Onshore	Spatio-temporal model	0.187	0.143	1.62e-04

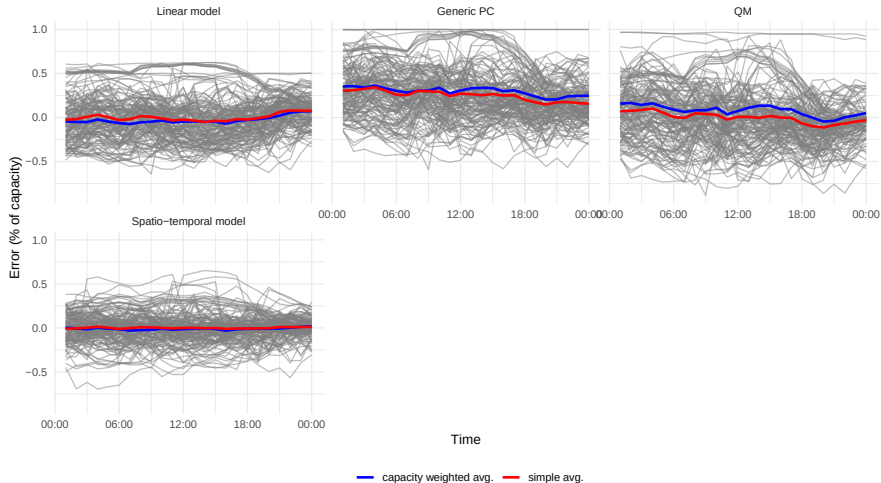
Estimates by hour

Power estimates Time Series 2024-08-10



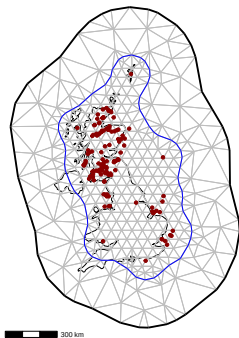
Errors by hour

Power estimates Time Series 2024-08-10

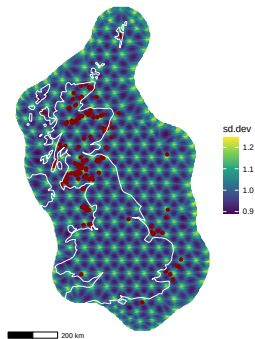


Model Mesh

Mesh used for spatial calibration model



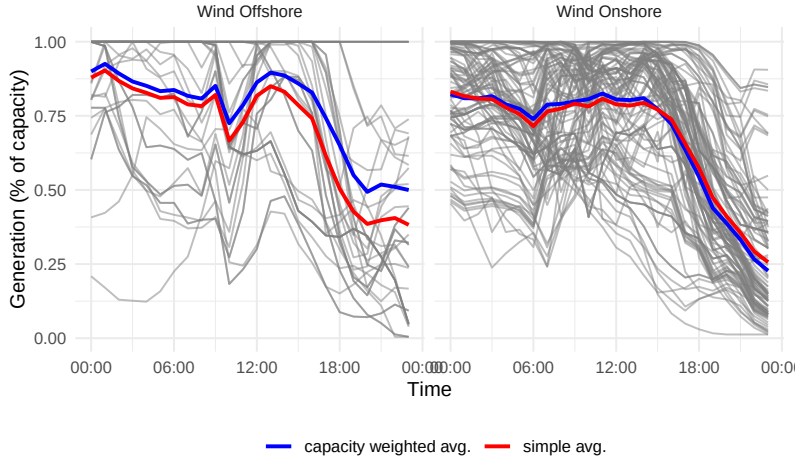
Mesh assessment



Time series of data and generic PC estimate

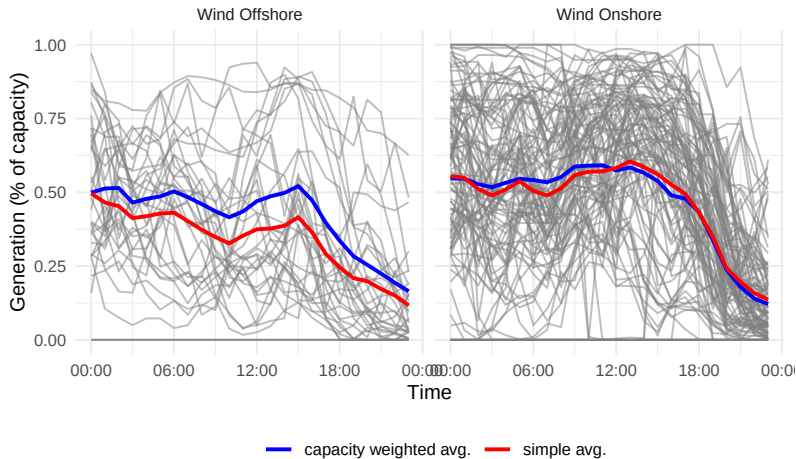
ERA5 data and generic PC estimate

ERA5 + Generic PC Time Series 2024-08-10

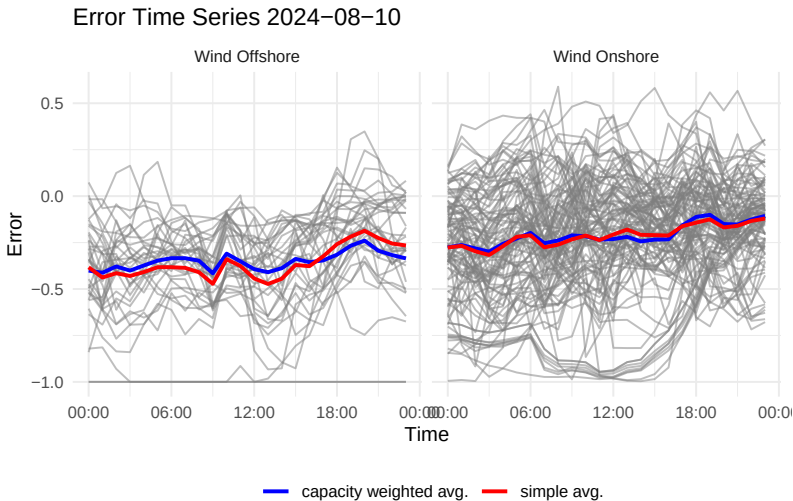


Observed data

Observed power Time Series 2024-08-10



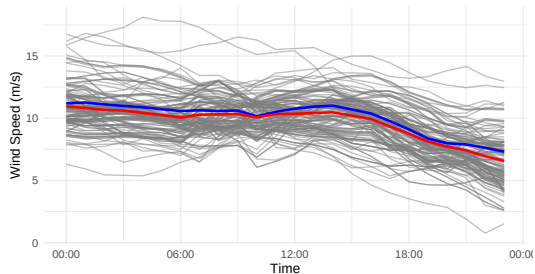
Error of generic PC estimate



ERA5 wind speed

Wind speed at wind farm locations

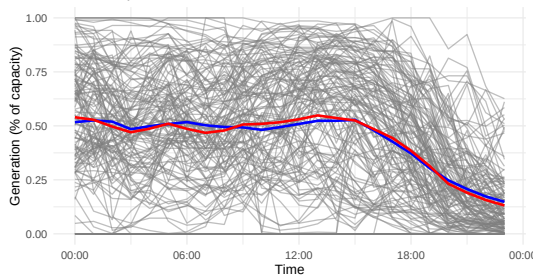
Wind Speed Time Series 2024-08-10



— capacity weighted avg. — simple avg.

Observed generation

Observed power Time Series 2024-08-10



— capacity weighted avg. — simple avg.

Appendix

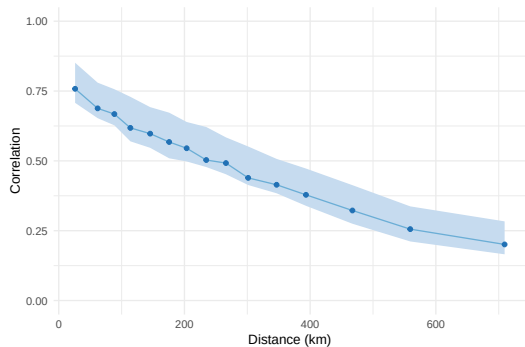
Project Goal

Project Goal

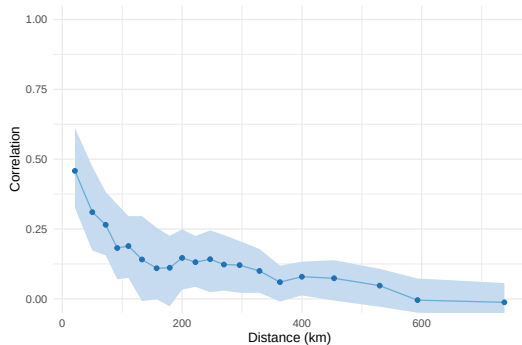
- Correct ERA5 wind power biases using observed generation data.
- Reduce error metrics (RMSE/MAE/bias).
- Ensure spatially consistent estimates that generalise out-of-sample.
- Capture low-wind events: frequency, duration, and spatial extent.

Spatial correlation

Correlation for Elexon generation



Correlation for error after calibration



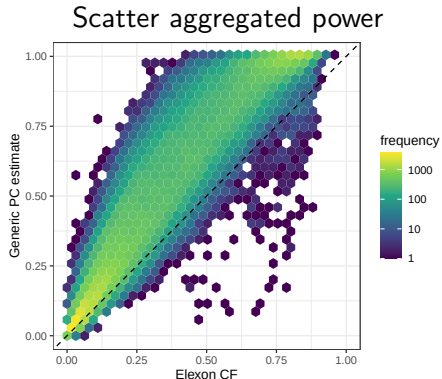
Calibration of generic power curve estimate

GB level aggregation

- Nearest ERA5 grid for wind speed
- Vertical interpolation using a power law
- Generic Power Curve (onshore, offshore)
- Normalised power aggregated at GB level

$$p_{\text{tot},t} = \sum_i p_{it} / \sum_i C_i$$

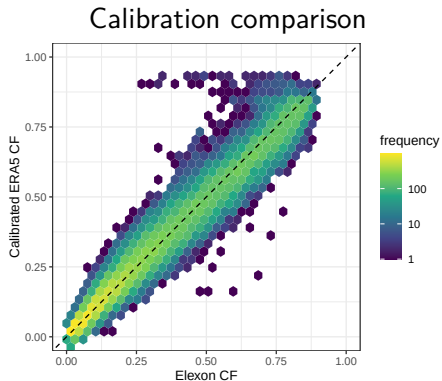
- Hourly resolution
- Potential generation (ex curtailment)



Linear model

- Using step AIC criteria selection:
 - k : Type (offshore / onshore)
 - m : month, h : hour
 - w_t : wind speed

$$p_t^{calib} = \alpha_{k,m,h} + \beta_{k,m} \hat{p}_t + \sum_{j=1}^3 \phi_j w_t^3$$

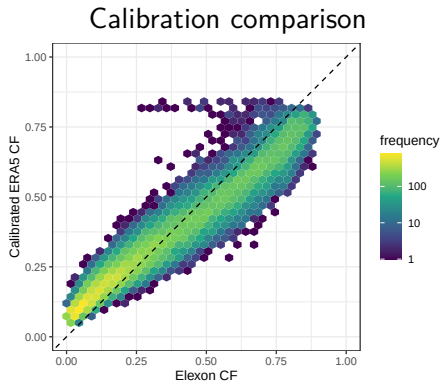


Autoregressive error

- Using step AIC criteria selection:
 - k : Type (offshore / onshore)
 - m : month, h : hour
 - w_t : wind speed
 - u_t : AR1

$$p_t^{calib} = \alpha_0 + \sum \alpha_e + \beta_k \hat{p}_t + f_w(w_t) + u_t$$

where α_0 is the intercept, the α_e $e \in k, m, h$ represent random effects (cyclic for month and hour), β_k is a random slope, f_w is smooth effect (RW2).

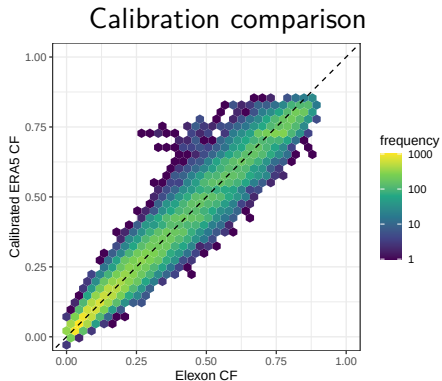


Hierarchical Bayesian model

- Turning multiplicative corrections into smooth functions

$$p_t^{calib} = \alpha + \beta_k + \beta_m + \beta_h + s_p(p_t) + s_w(w_t)$$

where β_k , β_m , β_h are random effects and s_p and s_w are smooth functions

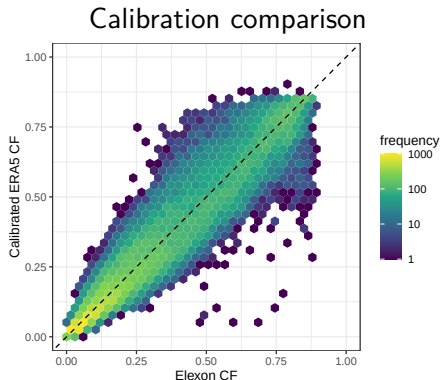


Quantile Mapping

- Quantile mapping aims to replicate observed distribution

$$p_t^{calib} = F_{obs}(F_{mod}^{-1}(p_t))$$

where F_{mod} and F_{obs} are the CDF of the power estimate and observed values



Comparison scores

model	RMSE	MAE	Bias
Generic PC	0.200	0.164	1.63e-01
AR inlabru	0.092	0.077	3.16e-04
AR1	0.075	0.062	8.57e-05
QM	0.075	0.054	-2.60e-05
Linear	0.052	0.038	1.39e-19
Linear inlabru	0.049	0.036	-2.61e-10

Power curve modelling

Power curve modelling

Components

- Power curve estimate from data
- Probability of zero generation
- Penalisation to make power curve resemble manufacturer's PC

Using a three-likelihood approach for power curve estimation:

- **1st likelihood:** Observed power is modelled as a smooth function of wind speed
- **2nd likelihood:** Zero inflated component
- **3rd likelihood** Penalisation towards a generic PC

Power model overview

- 1 $\pi_{it} = \pi(w, i) = \Pr(P_{it} = 0)$ depends of wind speed and location

$$L^{(0)}(\theta^0 | \mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_t [\pi_i(w_{it}, i)^{\mathbb{1}(P_{it}=0)} \times (1 - \pi_i(w_{it}, i))^{\mathbb{1}(P_{it}>0)}]$$

- 2 Generation when $P_{it} > 0$ also depends on wind speed / location. Learned as a smooth function from data: $s^{\text{pc}}(w, i) \in (0, 1)$

$$s^{\text{pc}}(w_{it}, i) = \mathbb{E}(P_{it} | P_{it} > 0, w_{it}, i) = \mu_{it}^+ \\ L^{(\beta)}(\theta^{\text{pc}} | \mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_{t: P_{it} > 0} \text{Beta}(P_{it} | \mu_{it}^+, \phi),$$

where θ^0 and θ^{pc} represents the parameters of the smooth functions, $\pi(w, i)$ and s^{pc} , and ϕ the precision of the Beta distribution.

Penalisation component

- 3 Finally, we penalise s^{pc} to be close to our generic power curve estimate, $\hat{P}\hat{C}_i(w)$: $d_{it} = |\hat{P}\hat{C}_i(w_{it}) - s^{\text{pc}}(w_{it}, i)|$ should be small.
- Or maximise the negative of $\tau_{\text{ps}} \sum d_{it}^2$ for a certain penalisation parameter τ_{ps} .
- Adding that penalty term is similar to assume $d_{it} \sim N(0, 1/\tau_{\text{ps}}^{-1})$

$$\max_{\theta} \quad l^{(0)}(\theta) + l^{(\beta)}(\theta) - \tau_{\text{ps}} \sum_{i,t} d_{it}^2$$

where $\theta = (\theta^0, \theta^{\text{pc}}, \tau_{\text{ps}})$ represents the parameters of both likelihoods, plus the penalisation parameter.

Power curve model equations

Model is a Zero Inflated Beta (ZIB) with penalisation component:

Zero-probability model (Bernoulli)

$$\eta_{it}^{(0)} = \alpha_{\text{zero}} + f_{\text{zero}}(w_{it}, i),$$

$$Z_{it} \mid \eta_{it}^{(0)} \sim \text{Bernoulli}(\pi_{it}), \quad \pi_{it} = \frac{1}{1 + \exp\left\{-\eta_{it}^{(0)}\right\}},$$

Positive-output model (Beta)

$$\eta_{it}^{(\beta)} = \alpha_{\text{pc}} + f_{\text{pc}}(w_{it}, i),$$

$$P_{it}^{\text{obs}} \mid \eta_{it}^{(\beta)} \sim \text{Beta}(\mu_{it}^+, \phi), \quad \mu_{it}^+ = \frac{1}{1 + \exp\left\{-\eta_{it}^{(\beta)}\right\}},$$

Pseudo-likelihood model (Gaussian)

$$\hat{P}C_i(w_{it}) \mid \eta_{it}^{(\beta)} \sim \mathcal{N}\left(\eta_{it}^{(\beta)}, \tau_{\text{ps}}^{-1}\right), \quad \tau_{\text{ps}} = \text{precision or penalisation}.$$

Details on smooth components

- The smooth components $f_{\text{zero}}(w, i)$ $f_{\text{pc}}(w, i)$ is modelled as a 1D SPDE over wind speed with a uniform correlation matrix across locations $i = 1, \dots, I$.

$$f(w, i) = \sum_{k=1}^I \beta_k \psi_k(w),$$

where $\psi_k(w)$ are basis functions defined over wind speed and β_k are coefficients. The correlation structure of $f(w, i)$ assumes a Matern covariance function: Let w_s, w_t represent two wind speed values in location i , and $d = |w_s - w_t|$ their distance, then:

$$\begin{aligned} \text{Cor}(d) &= \text{Cor}(f(w_s, i), f(w_t, i)) \\ &= \frac{1}{2^{\nu-1} \Gamma(\nu)} (\kappa d)^\nu K_\nu(\kappa d), \quad \alpha = \nu + d/2, \end{aligned}$$

Details on smooth components

- Additionally a uniform correlation matrix is assumed across locations $j, k \in \{1, \dots, I\}$. That is:

$$\text{Cor}(f(w, j), f(w, k)) = \rho, \quad j \neq k$$

which helps to pool information across locations, benefiting locations with less data.

Power estimate

This leads to the following power estimate at time t and location i :

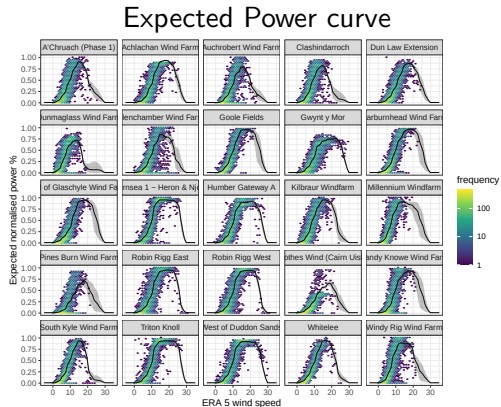
$$\hat{P}_{it} = \begin{cases} 0, & \text{with probability } \pi_{it}, \\ \mu_{it}^+, & \text{with probability } 1 - \pi_{it}. \end{cases}$$

Which leads to the expected power estimate:

$$E[P_{it}|w_{it}, i] = (1 - \pi_{it}) \times \mu_{it}^+$$

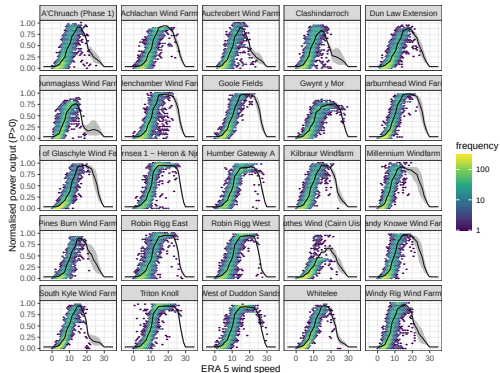
Power curve model estimates

- Starting models with
 - 1Y of data (2024)
 - 25 wind farms
 - Hourly data
 - Excluding outages listed in REMIT

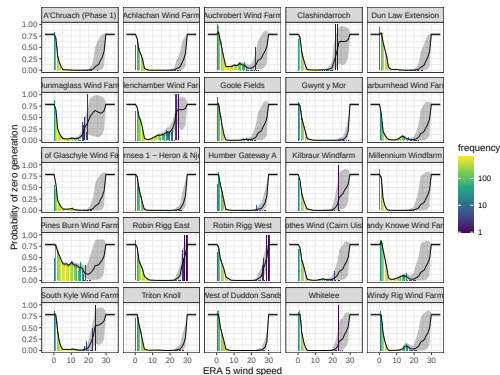


Power curve model estimates

Power curve estimate



Probability of zero generation

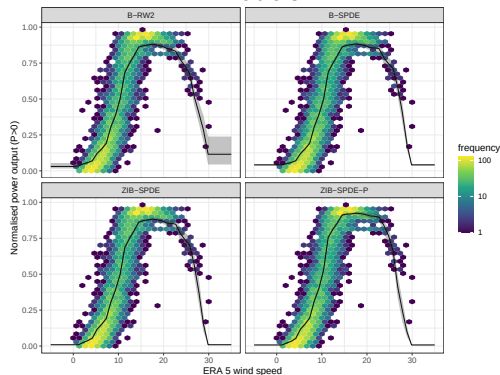


Alternative versions

- Beta model with RW2 power curve (B-RW2)
- Beta model with 1D SPDE power curve (B-SPDE)
- ZIB model with 1D SPDE power curve (ZIB-SPDE)
- ZIB model with 1D SPDE power curve penalised (ZIB-SPDE-P)

Zero Inflated models better capture low wind power regions.

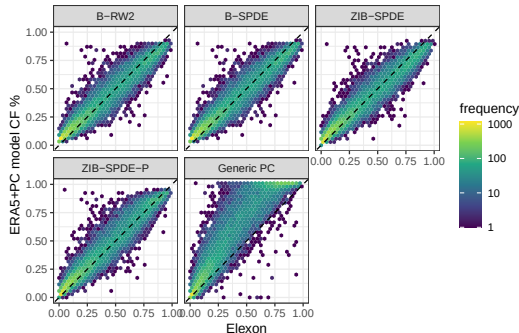
Power curve estimates under different models



ZIB model vs observed data

- Power curve modelling highly improves the calibration
- ZIB models deal better with the tail regions
- Penalised version actually pulls estimate toward Generic PC
- We could pull estimate instead to a 5 parameter curve estimate

Power curve models vs observed data



Error metrics

Error metrics calculated on 30% of the sampled wind farms data.

model	RMSE	MAE	Bias
Generic PC	0.2188	0.1583	0.1469
B-RW2	0.0995	0.0723	0.0191
B-SPDE	0.0994	0.0722	0.0190
ZIB-SPDE	0.0976	0.0691	0.0103
ZIB-SPDE-P	0.1016	0.0694	0.0269

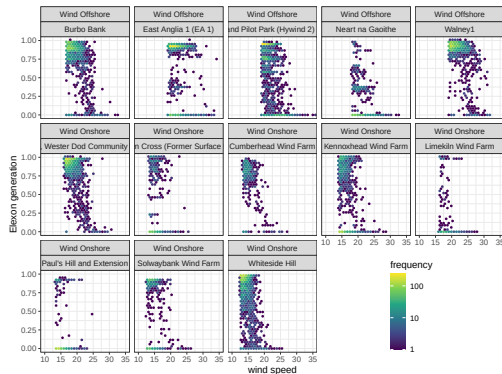
Pooling data for cut-off estimation

- Not enough data to estimate at wind farm level
- Even using all 5 years of data (2019-2024), only a few farms show cut-off behaviour

Pooling data at high wind speeds

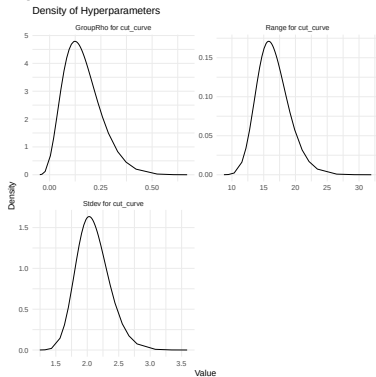
wind speed	prop. $P=0$	prop. $P \leq 0.02$	n obs.
[0,15]	0.22	0.23	3292
(15,20]	0.13	0.14	7062
(20,25]	0.11	0.13	5266
(25,30]	0.43	0.48	670
(30,Inf]	0.76	0.79	63

Tail behaviour of power curve

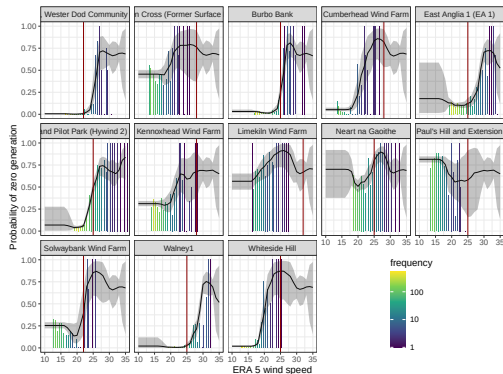


Model to estimate cut-off by wind speed

Hyperparameters of the cut-off model



Probability of zero generation at high wind speeds



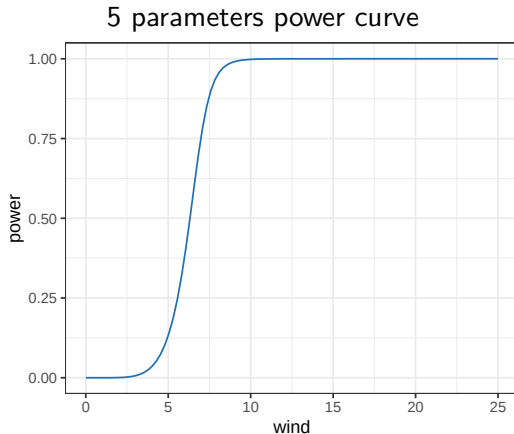
Power Curve fitting

Classic parametric shape

The five-parameter formula (Lydia et.al, 2013):

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

where: A and D are the upper and lower asymptotes, C controls the inflection point, B is related to the slope at inflection point, and G controls the asymmetry.



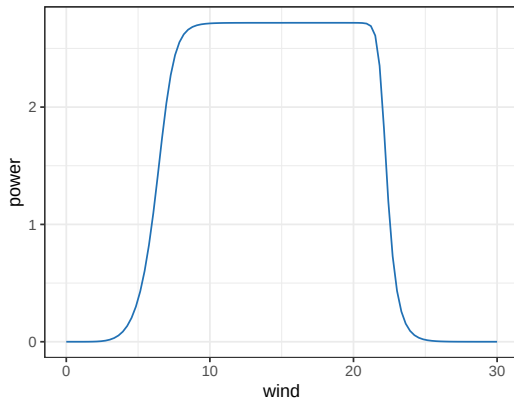
Parametric shape for cut-off

Adding extra parameters we can model the decay after cut-off:

$$P(u) = D + \frac{A - D}{(1 + (u/C_1)^{B_1})^G (1 + (u/C_2)^{B_2})^G}$$

where: A and D are the upper and lower asymptotes, C_1 is the first inflection point, C_2 is related to the cut-off, B_1 is related to the slope at inflection point, B_2 to the slope during cut-off, and G controls the asymmetry.

7 parameters power curve



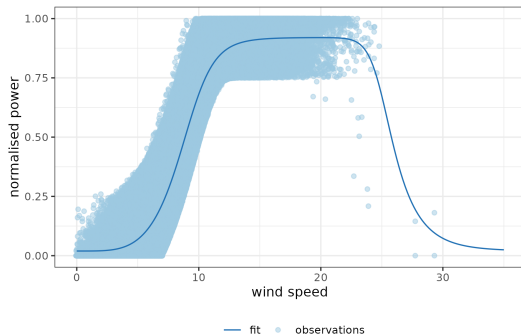
Parametric fit for Scottish wind farms

- We can fit the parametric shape using optim
- As long as there is a reasonable amount of points showing the cut-off optim can converge

Estimates from data

par	estimate
A	0.92
B1	-9.74
C1	10.06
D	0.02
G	0.43
B2	35.64
C2	24.92

Data and power curve estimate



Next Steps

- Spatiotemporal calibration model
- Non-stationary covariance model
- Comparison against benchmarks

Data sources

Wind speed

- ERA5 at wind farms
 - Hourly data
 - Spatial resolution $0.25^{\circ} \times 0.25^{\circ}$
 - 10m and 100m heights

Wind power

- Elexon BMU data (since 2019)
 - Half hourly data
 - Generation, curtailment, potential, capacity
 - Outage data (REMIT)
- REPD database
 - Location, turbine height, capacity

Previous research

Power curve modelling

Binning method

$$P_i = \frac{1}{n_i} \sum_{j=1}^{n_i} P_{ij}$$

where: P_{ij} is the j th power observation in bin i and n_i no. of observations in bin i

Logistic

$$P(u) = a \frac{1 + m \exp(-u/\tau)}{1 + n \exp(-u/\tau)}$$

where a represents the upper asymptote, n, m shape the lower asymptote, and τ controls the transition.

Power curve modelling

5 parameter curve

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

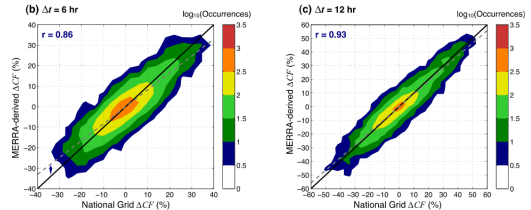
where: A and D are the upper and lower asymptotes, C is the inflection point, B the slope at inflection point, and G controls the asymmetry.

Reanalysis data to quantify extreme wind power statistics

D. Cannon, D. Brayshaw, et. al (2015)

- MERRA wind speed validated with MIDAS
- Vertical interpolation with a logarithmic change
- Calibrated power curves based on manufacturers PC
- Use that to analyse extreme low and high levels, and ramps

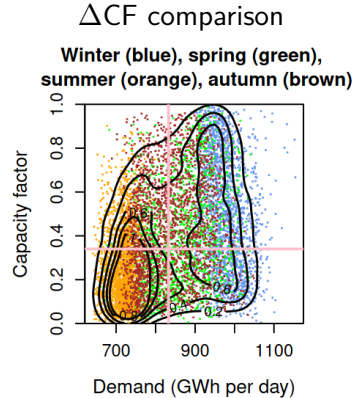
ΔCF comparison



Balancing energy

H. Thornton, D. Brayshaw (2020) analyse the relationship between weather, energy demand, and wind power.

- ERA Interim wind speed
- Cubic power curve with air density correction
- CF compared with GB average from other studies
- Seasonal effects on Demand and

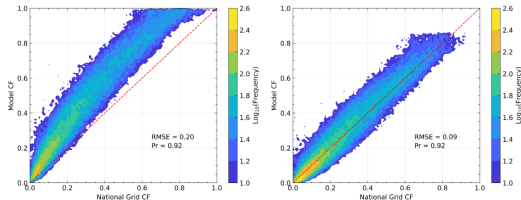


Analysis of extreme wind droughts

Panit Potisomporn, C. Vogel (2024) perform an extreme value analysis of wind droughts in GB.

- Use ERA5 wind speeds calibrated to MIDAS with QM
- Build a ML algorithm that learns how to extrapolate wind speed from 10m to hub height
- Use a 5 parameter logistic function to model power curve
- Model energy losses with factors by type.

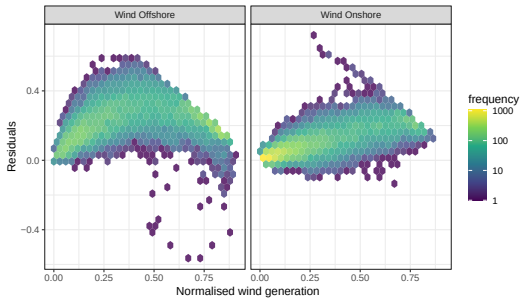
CF calibration



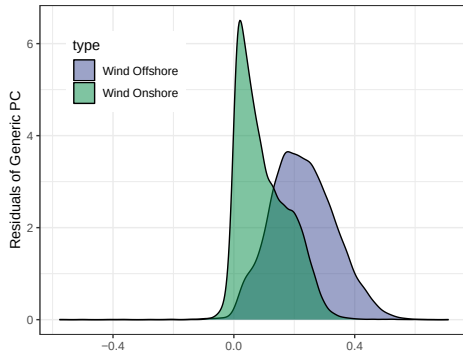
ERA5 + generic power curve error analysis

Error by Onshore / Offshore

Errors by observed power

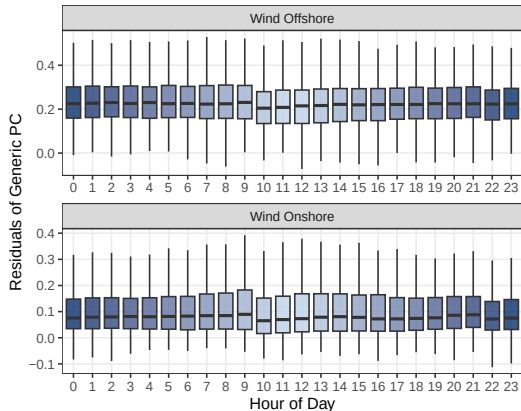


Error density

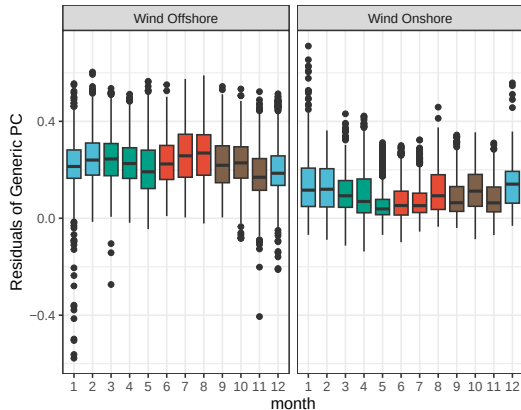


Errors through time

Error by hour



Error by month



Challenges in spatial wind calibration

Challenges in spatial wind calibration

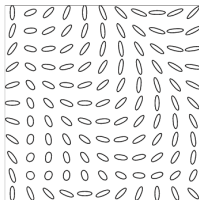
- Non-stationarity
 - (space) spatially varying coefficients
 - (wind direction) anisotropy
 - (time) regimes / storms and other weather events
- Physically informed model
 - Regularisation / penalisation
- Data dimensionality reduction
 - Aggregate time dimension (day)

Non-stationarity approaches found in the literature

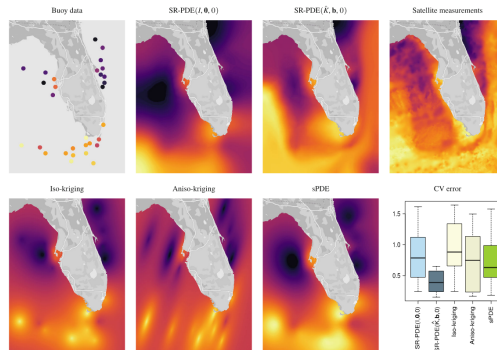
Regression with PDE regularisation

- Tomasetto et al.(2024) model anisotropy through a physics informed regression model
- Including physical knowledge through a PDE regularisation term

Anisotropic non-stationary diffusion



Sea Surface Temperature (SST)



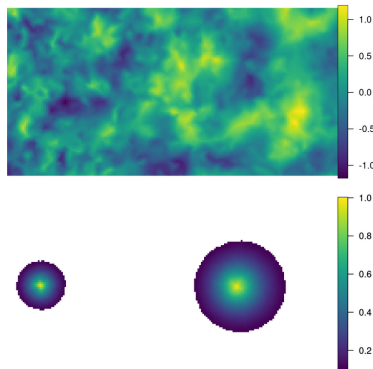
Explanatory variables in the covariance

- Ingebrigtsen, Lindgren, and Steinsland (2014)
- Regression on ranke and precision parameters of spatial term

$$\log \tau(\mathbf{s}) = b_0^{(\tau)}(\mathbf{s}) + \sum_{k=1}^p b_k^{(\tau)}(\mathbf{s}) \theta_k, \quad (2)$$

$$\log \kappa(\mathbf{s}) = b_0^{(\kappa)}(\mathbf{s}) + \sum_{k=1}^p b_k^{(\kappa)}(\mathbf{s}) \theta_k. \quad (3)$$

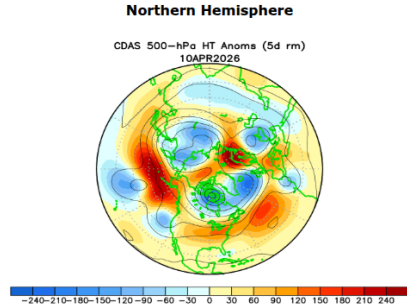
Field and correlation at two points



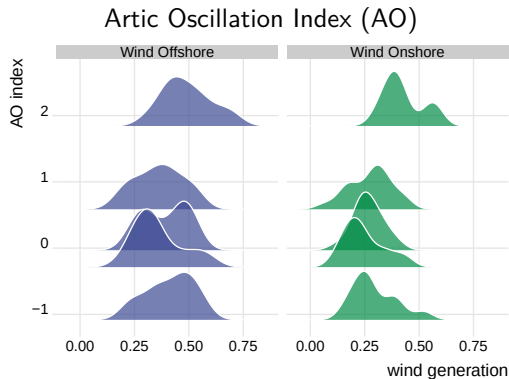
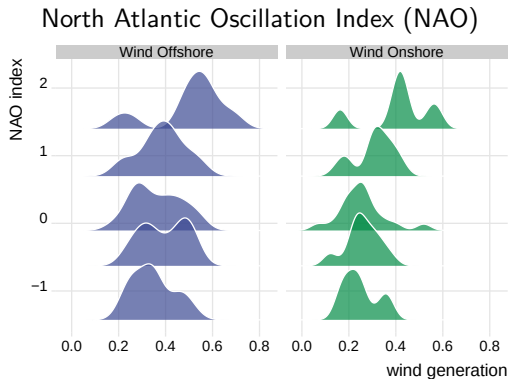
Variables relevant to non-stationarity in wind

- Wind dynamics are non-stationary in space and time
- Weather regimes, storms, other events
- Circulation patterns (NAO, AO, EA, SCAN)
- NAO + leads to higher wind speeds UK and more variability

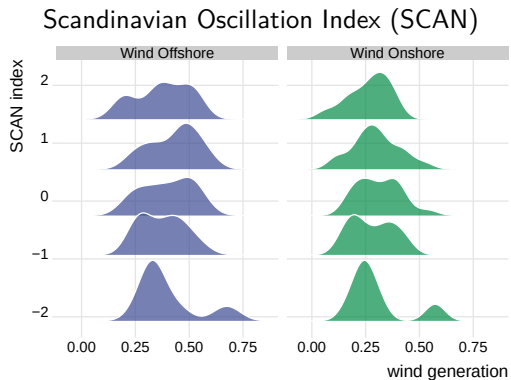
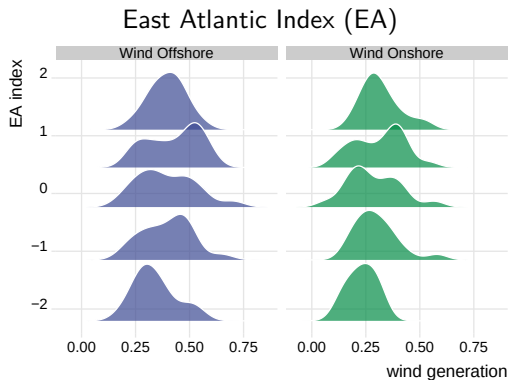
North Atlantic Oscillation (NAO) *500-hPa Height Anomalies*



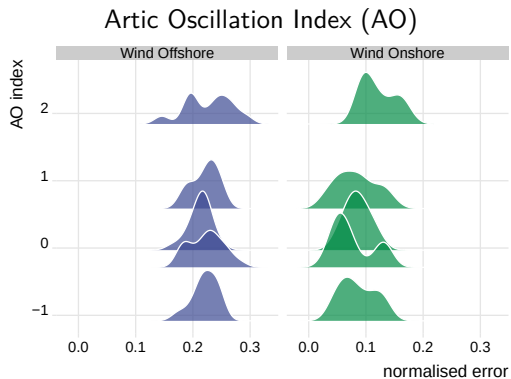
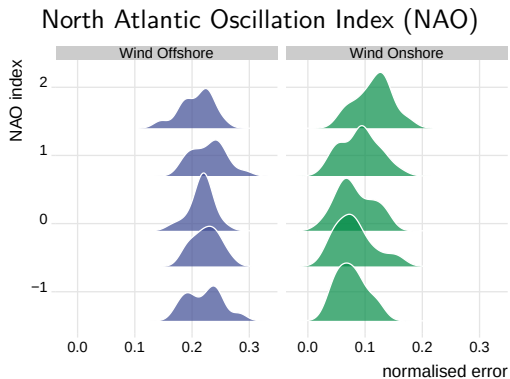
Circulation patterns and wind power



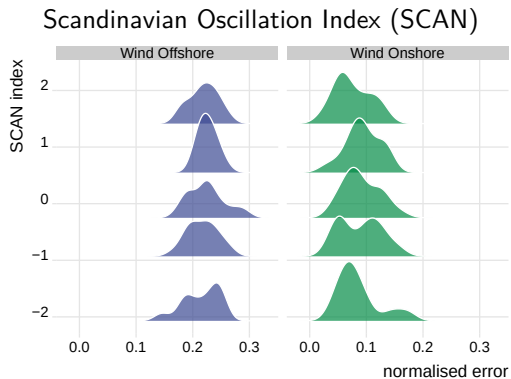
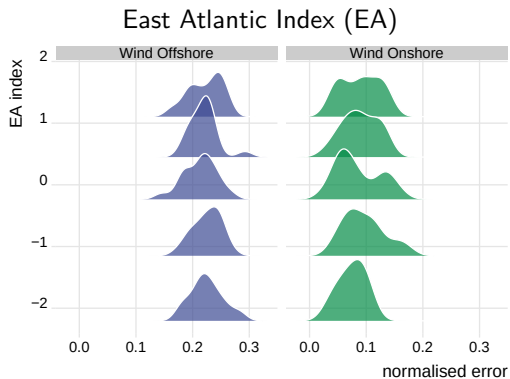
Circulation patterns and wind power



Circulation patterns and error in wind power estimates



Circulation patterns and error in wind power estimates



References

- Cannon, D. J., D. J. Brayshaw, J. Methven, P. J. Coker, and D. Lenaghan. 2015. "Using Reanalysis Data to Quantify Extreme Wind Power Generation Statistics: A 33 Year Case Study in Great Britain." *Renewable Energy* 75 (March): 767–78. <https://centaur.reading.ac.uk/38448/>.
- Potisomporn, P. 2024. "Spatiotemporal Variability of Extreme Low Wind Power Events in Great Britain - ORA - Oxford University Research Archive." [Http://purl.org/dc/dcmitype/Text](http://purl.org/dc/dcmitype/Text).
<https://ora.ox.ac.uk/objects/uuid:de37bee2-1813-42e5-a69c-9b15f038d096>.
- Thornton, Hazel E. 2020. "Atmospheric Circulation, Seasonal Predictability and Britain's Energy System." PhD thesis, University of Reading.
<https://doi.org/10.48683/1926.00095832>.